QUANTUM FLOW TECHNOLOGIES

Quantum-safe by design

PRODUCT VISION & ROADMAP · OCTOBER 2026

Securing the World's Cryptography for the Quantum Era

Building Quantum Flow's PQC Discovery & Migration Platform: founder's vision, product strategy and a four-month execution roadmap.

PREPARED BY

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AUDIENCE

Quantum Flow team, partners and prospective investors

STATUS

Version 1.0 · Living document · Confidential

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> *This is a living document. It sets our shared direction rather than a frozen specification; we will refine priorities, scope and architecture as we learn from testing, customers and partners. Targets are stated as targets, not guarantees.*

FROM THE FOUNDER

A note from the Founder & CEO

Every organisation that relies on digital trust will have to change its cryptography. We intend to be the company that shows them how.

To our team, our partners and those who may invest in this journey,

Quantum computing will reshape the foundations of digital security. The public-key cryptography that protects banking, government, telecommunications and critical infrastructure today was not designed for that future. The transition to post-quantum cryptography will be one of the largest security migrations the world has undertaken, and it begins with a deceptively simple question that most enterprises cannot yet answer: **where, exactly, is our cryptography, and what depends on it?**

Quantum Flow is building the platform that answers that question and then guides the journey that follows. Our ambition is large and deliberate: to build a globally competitive quantum-safe cybersecurity company, built in India and designed for the world, and to work towards a valuation of **US$1 billion by March–April 2027** with a presence in **at least ten countries**. This is a stretch goal, not a forecast. It will be earned through proprietary IP, credible customers, disciplined execution and investor confidence.

This journey is already under way. I am in active discussions with people and potential partners across the UAE, Vietnam, the United States and Europe, and pursuing strategic relationships in India. Drawing on my experience of building deep-technology businesses, engaging investors and creating enterprise value, I have secured **investment commitments from two global investors/funds** for this venture. The funding pathway is linked to our next milestone, a credible beta and pilot-deployable version of our PQC discovery scanner and migration platform, together with the agreed investment conditions and definitive documentation.

Our public-sector work matters greatly. Based on the confirmation I have received from the Government of Karnataka, we are working towards adoption of this platform as a national reference architecture, with scope, approvals, implementation responsibilities and formal documentation still to be completed. I have also been engaging with, and receiving encouragement from, senior figures in India's quantum and technology ecosystem, including Dr. Ajai Chowdhry, as well as senior state leadership. These relationships open doors; delivery is what will keep them open.

Our approach is clear. We will not invent cryptography. We will build on published standards and proven components, and create our own IP where it matters most: discovery, the cryptographic asset model, risk analysis, migration orchestration, policy controls, integrations and the evidence trail that proves the work was done right. We will deliver this roadmap by organising the capable team we already have around sharp priorities.

The next three to four months could be a turning point for Quantum Flow and for every person building this capability. Few professionals anywhere have hands-on experience of cryptographic discovery, crypto-agility and PQC migration. Those who build it now will hold rare and valuable expertise. I want them to be recognised for it, and, as the business grows, rewarded fairly through transparent and properly documented company processes.

Our ambition is global. Our execution will be disciplined, measurable, honest about limitations and collaborative. Let us build something the world will need.

Nitin M. Jakaty

Founder & CEO, Quantum Flow Technologies

EXECUTIVE SUMMARY

At a glance

A standards-based platform that finds an organisation's cryptography, ranks the quantum risk and orchestrates a safe, evidenced migration.

4
months to a pilot-ready platform and investor demo

2
integrated workstreams: discovery scanner and migration platform

3
NIST post-quantum standards at the core: FIPS 203, 204, 205

10+
countries targeted for presence by 2027

How the platform works

01 **Discover** — Find certificates, keys, protocols and crypto in code

02 **Inventory** — Build an evidence-backed CBOM

03 **Prioritise** — Score risk by exposure and data lifetime

04 **Plan** — Dependency-aware migration packages

05 **Migrate** — Controlled PQC and hybrid changes

06 **Assure** — Test, verify and keep the audit trail

The problem

Organisations rely on RSA and elliptic-curve cryptography that a sufficiently capable quantum computer is expected to break. Most cannot see where it is used, what depends on it, or where to begin.

Our answer

One platform, one asset model: discovery feeds inventory, inventory feeds risk prioritisation, and prioritisation drives a controlled, auditable migration.

Our defensible IP

Not the algorithms, which are public standards, but the software and know-how to discover, connect, prioritise, migrate and continuously assure cryptographic systems.

Route to market

A Cryptographic Readiness Assessment that opens the door, followed by recurring platform licences and managed cryptographic-posture services.

WHERE WE STAND

Investment commitments from two global investors/funds, linked to a beta and pilot-deployable product and the agreed conditions and documentation.

Active market discussions across India, the UAE, Vietnam, the United States and Europe, and work towards Government of Karnataka adoption as a national reference architecture, subject to formal approvals.

SECTION 01

Why now: the quantum-safe imperative

The standards are published, the policy direction is set, and the clock started long before the first cryptographically relevant quantum computer will arrive.

Harvest now, decrypt later

Adversaries can record encrypted data today and decrypt it once quantum capability matures. Any data that must stay confidential for years is already exposed.

The standards have arrived

In August 2024 NIST published its first post-quantum standards: ML-KEM (FIPS 203), ML-DSA (FIPS 204) and SLH-DSA (FIPS 205). Organisations now have a defined destination.

Policy is moving in India

The Department of Science and Technology's February 2026 report on implementing a quantum-safe ecosystem in India sets direction on PQC migration, testing, certification and adoption.

Migration takes years

Cryptography is embedded in applications, devices, vendors and certificates. Large enterprises will need multi-year programmes, and every one of them starts with discovery.

This is why discovery and inventory are the first, unavoidable step of every quantum-safe programme, and why we have chosen them as our entry point. An organisation cannot migrate what it cannot see, and it cannot prioritise without evidence. A platform that delivers that visibility, and then carries it through to a controlled migration, sits at the heart of a market that every regulated sector will have to enter.

SECTION 02

Vision and objectives

Our own standards-based platform to help organisations discover their cryptography, understand quantum exposure and migrate safely to post-quantum cryptography.

What we are building towards

An IP-owning product company

A globally competitive player in quantum-safe cybersecurity, built on our own software and know-how.

A complete readiness platform

Cryptographic discovery, inventory, risk prioritisation and migration orchestration in one coherent product.

A trusted partner

For enterprises, public-sector organisations and critical infrastructure, with an architecture built for international deployment.

A recurring-revenue business

Subscriptions, enterprise licences, private deployments and managed assessments.

An ecosystem contributor

Strengthening India's quantum-safe ecosystem through standards-aligned implementation, testing, interoperability and indigenous software IP.

A global footprint

Repeatable deployment across markets, with localisation, compliance and deployment adapted country by country.

Milestones we are working towards

These are targets, not guarantees. A prototype is not production readiness, certification or broad compatibility, and we will always describe our maturity accurately.

HORIZON	TARGET OUTCOME	EVIDENCE OF COMPLETION
First 2 weeks	Scope, owners, threat model and architecture	Product brief, backlog and test plan
By week 4	Discovery prototype	Demo of TLS endpoints, certificates and structured findings
By week 8	Scanner MVP	Release candidate, coverage matrix and test results
By month 3	Inventory / CBOM and migration plan	Sample report and dependency-aware remediation plan
By month 4	Controlled PQC / hybrid migration demo	Interoperability, performance, regression and rollback evidence
By month 4	Pilot agreed or under way, where customer access permits	Written scope, permissions and success criteria

SECTION 03

Product strategy: our IP, built on open standards

We do not reinvent cryptography. We build the intelligence around it.

Quantum Flow will not invent its own cryptographic algorithms. We implement and integrate published standards through maintained libraries and supported protocols, subject to licensing, security and compatibility review. Our IP sits above and around those standards: discovering cryptographic assets, connecting evidence, building a Cryptographic Bill of Materials (CBOM), ranking risk, generating migration plans, coordinating approved changes, verifying results and maintaining an audit trail.

The platform, layer by layer

LAYER	WHAT WE BUILD OR INTEGRATE	IP PRINCIPLE
Discovery engine	Network and TLS observation, certificate collection, source and dependency scanning, configuration import	Our own orchestration, normalisation and evidence model
Cryptographic inventory / CBOM	Asset records, relationships, confidence, ownership, location and evidence	Our own data model and asset graph
Risk prioritisation	Explainable rules for exposure, data lifetime, business criticality and migration complexity	Our own configurable rules and methodology
Migration planning	Dependency-aware work packages, recommended sequencing and approvals	Our own workflow and policy engine
Migration execution	Adapters for supported TLS, library and application configurations, with tests and rollback	Built on established primitives and protocols
Assurance and reporting	Before-and-after comparison, test evidence, audit history and exceptions	Our own evidence and reporting layer
Cryptographic algorithms	ML-KEM, ML-DSA, SLH-DSA and other standards where appropriate	Standards we implement, never claim to own

OUR DEFENSIBLE ADVANTAGE

Our IP is not the fact that NIST publishes ML-KEM or ML-DSA. It is the software and know-how we create to discover, connect, prioritise, migrate and continuously assure cryptographic systems, at enterprise scale and with evidence.

SECTION 04

Market landscape and partnerships

A fast-forming category, where focus, evidence and partnership will matter as much as breadth.

The quantum-safe category is taking shape quickly. Established players such as QNu Labs, whose QShield 2.0 platform spans discovery, risk assessment, crypto-agility, PKI, key management and AI security, demonstrate both the breadth of the opportunity and the language customers are beginning to use.

We study the public market to understand customer needs, user journeys and gaps that Quantum Flow can address. We build our product independently, from our own designs and code, and we respect the proprietary designs, content and confidential information of others. Where there is mutual value, we will pursue partnership, interoperability or commercial relationships professionally and openly.

Our focus

Our first release is intentionally focused: **discovery, risk prioritisation and a controlled migration demonstrator**. Broader capabilities, such as PKI, vault and key management, quantum entropy, AI security and national-assurance services, will follow once customers have validated the core. Focus is how a young company wins a category.

Public reference:
QNu Labs QShield 2.0 product page

SECTION 05

Standards foundation

Official standards and guidance are our baseline library. We build against them, test against them and track every revision.

A named owner maintains a standards register and checks for errata, revisions and new guidance at least monthly and before every release.

RESOURCE	PURPOSE	OFFICIAL SOURCE
NIST FIPS 203 (ML-KEM)	Key-encapsulation mechanism for establishing a shared secret	csrc.nist.gov/pubs/fips/203/final
NIST FIPS 204 (ML-DSA)	Post-quantum digital signatures	csrc.nist.gov/pubs/fips/204/final
NIST FIPS 205 (SLH-DSA)	Stateless hash-based post-quantum signatures	csrc.nist.gov/pubs/fips/205/final
NIST PQC project	Standards and project status	csrc.nist.gov/projects/post-quantum-cryptography
NIST CSRC publications	Transition guidance, test publications and errata	csrc.nist.gov/publications
NIST SP 800-57 Part 1	General key-management guidance	csrc.nist.gov/pubs/sp/800/57/pt1/r5/final

RESOURCE	PURPOSE	OFFICIAL SOURCE
DST: Quantum Safe Ecosystem in India	Indian policy direction on PQC migration, testing, certification and adoption	dst.gov.in (report PDF)
National Quantum Mission	Mission overview and official programme links	dst.gov.in/national-quantum-mission-nqm
NQM start-up support guidelines	Eligibility, ownership, IP and application requirements	dst.gov.in (guidelines PDF)

Putting the NIST standards to work

Not every engineer needs to read every standard cover to cover. The technical lead prepares a short implementation note for each standard and points the team to the sections that matter. Our method:

1. **Source and record.** Download FIPS 203, 204 and 205 from the official NIST pages; record publication date, errata status and the date checked.
2. **Summarise.** One page per standard: purpose, parameter sets, input and output behaviour, key and signature sizes, implementation considerations, test vectors and interoperability constraints.
3. **Select maintained libraries.** Check licence, maintenance, platforms, security advisories and validation claims; record exact versions.
4. **Use supported APIs, never re-implement primitives.** Build a test harness using official or library-provided known-answer tests.
5. **Prove it in the lab.** Test success cases, malformed inputs, error handling, performance, memory, concurrency and failure modes.
6. **Hybrid done properly.** Use a supported protocol implementation and documented combination method; never a home-grown hybrid handshake.
7. **Claim precisely.** Distinguish algorithm standardisation, implementation validation and product certification. Using a NIST-standardised algorithm does not make a product “NIST certified”.
8. **Track change.** Review NIST errata and updates before every release.

> **INITIAL ALGORITHM PRIORITIES**
>
> ML-KEM for supported key-establishment use cases; ML-DSA for supported signature use cases; SLH-DSA as an additional signature option where appropriate. Support varies across TLS, PKI, HSM, operating-system and vendor stacks, so we demonstrate compatibility configuration by configuration.

The DST report is policy and ecosystem guidance, not a software framework. We translate its recommendations into product requirements and test plans, and verify the latest version of any document before relying on it in a customer or grant submission.

SECTION 06

Workstream A: the discovery scanner

Our first visible product milestone, delivered in eight weeks. It reports what it can actually observe, keeps evidence for every finding and states its coverage limits clearly.



WEEK 1 · DEFINE SCOPE AND BUILD THE LAB

- The product owner writes a two-page MVP brief: target user, use case, in-scope asset types, output format, out-of-scope areas and acceptance criteria.
- Build an isolated lab: Linux server, TLS-enabled web server, test certificate authority, sample application, source repository and container image, plus Windows where available.
- Seed known test assets: TLS versions and cipher suites, RSA and ECC certificates with varied expiry and algorithms, crypto-library references and deliberately outdated configurations.
- Scan only systems Quantum Flow owns or has explicit written permission to test; scope and approval come before any customer scan.
- Set up Git with branch protection, code review, issue tracking, CI pipeline, secrets handling and the development environment.

WEEKS 2–3 · IMPLEMENT THE FIRST COLLECTORS

- **Network/TLS collector:** for approved host and port targets, records TLS version, negotiated cipher suite, certificate chain, validity, signature algorithm and observable key-exchange details.
- **Certificate collector:** parses PEM/DER certificates for subject, issuer, serial, validity, public-key algorithm and size, signature algorithm and chain relationships.
- **Configuration importer:** starts with one or two approved export formats from servers, load balancers, proxies, cloud services or security tools.
- **Source and dependency scanner:** identifies crypto libraries, API calls, algorithm references and dependency manifests, treated as indicators rather than proof of runtime use.
- Every collector emits a common structured record: source, timestamp, target, method, confidence, evidence, collector version and error status.

WEEKS 3–5 · INVENTORY AND REPORTING

- Data structures for assets, observations, certificates, algorithms, applications, dependencies, owners, evidence and scan runs.
- Stable IDs and de-duplication, so repeat scans recognise the same item.
- Clear status for every record: observed, inferred, imported or unknown.
- Reports with summary, findings, evidence, confidence, business-impact fields, limitations and recommended next steps.
- Transparent rules: observable RSA/ECC use is flagged for PQC assessment, never labelled an exploitable vulnerability without context.

WEEKS 5–6 · RISK RULES AND USABILITY

- Configurable risk factors: data confidentiality lifetime, criticality, internet exposure, dependency complexity, vendor support and remediation urgency.
- Users can record business and application owner, vendor, environment and remediation date.
- Search and filter, CSV/JSON export and a readable HTML or PDF report.
- Every finding is explained with its supporting evidence; no opaque AI scoring in the MVP.

WEEKS 7–8 · TEST, PACKAGE AND DEMONSTRATE

- Compare scan results against the known ground-truth lab inventory.
- Test false positives, missed findings, malformed certificates, unreachable hosts, timeouts, duplicate scans, large inventories and permission failures.
- Publish a coverage matrix of supported protocols, file types, source languages and configuration formats.
- Document installation, permissions, data collected, storage, retention, access controls and uninstallation.
- Demonstrate the full journey: scan launch, inventory review, evidence inspection, prioritisation and report export.

> **SCANNER MVP ACCEPTANCE CRITERIA**
>
> Repeatable on the lab · evidence-backed findings · clear coverage limits · safe operation · reproducible tests · a working report · documented installation. A dashboard alone is not a completed scanner.

SECTION 07

Scanner architecture

Simple, testable components first; sophistication only where evidence demands it.

COMPONENT	PURPOSE	FIRST IMPLEMENTATION GUIDANCE
Scan controller	Creates scan jobs and enforces scope	Simple API and job queue, explicit allowlists and timeouts
Collectors	Observe TLS, certificates, code, dependencies and configuration	Separate modules with independent tests
Normalisation	Converts findings into a common schema	Versioned JSON schema and stable IDs
Inventory store	Stores assets, evidence, relationships and history	Relational database first; graph technology only if needed
Rules engine	Applies discovery and initial risk rules	Versioned, explainable and testable rules
API / UI	Runs scans, reviews assets and exports reports	Focused dashboard; correctness before polish
Security controls	Protects credentials, scope, tenant boundaries and logs	Least privilege, encryption, secrets vault and audit trail

Languages are chosen on team capability, packaging, performance and security. Python suits rapid collector development; Go or another compiled language suits a deployable agent. Third-party dependencies and licences are recorded from the very first commit.

SECTION 08

Workstream B: the migration platform

One platform, not two products. The scanner's asset model, evidence, ownership and risk data drive every migration decision.



Stage 1 · Migration planner

1. For each finding, identify the affected application, protocol, library, certificate, device or vendor product.
2. Record who controls the component: the customer, a third-party vendor, a cloud provider or an embedded device.
3. Map dependencies: what could break if the algorithm, certificate, library or protocol changed?
4. Classify remediation: configuration change, software upgrade, library update, certificate/PKI change, vendor action, hardware replacement or further investigation.
5. Generate a work package with sequence, prerequisites, tests, owner, approval, rollback and closure evidence.
6. Let the customer approve, defer or reject each recommendation, with a reason and a full audit trail.

Stage 2 · Controlled migration demonstrator

1. Select one supported application or TLS stack in the lab; universal enterprise migration is not the goal.
2. Establish a classical baseline: connectivity, certificate chain, latency, resource use and application behaviour.
3. Use a maintained implementation that supports the selected PQC algorithm and protocol configuration.
4. Where supported, test a documented hybrid key-establishment configuration, confirming client and server support.
5. Automate change in staging or a disposable environment; production changes always require customer approval.
6. Run functional, interoperability, load, performance, negative and regression tests.
7. Test rollback and verify the known-good state is restored after failure.
8. Record before-and-after configuration, library versions, tests, approvals and exceptions.
9. Publish the result: what changed, what passed, what remains unresolved and what is not supported.

Stage 3 · Crypto-agility

- A policy and configuration layer for approved algorithm profiles and protocol options.
- Cryptographic implementation kept behind supported libraries and stable interfaces wherever practical.
- Tracking of which applications support each profile and which need code or vendor changes.
- Versioned migration plans and policies, with approval required for high-impact changes.
- Maintained compatibility matrices and regression suites.
- Designed for on-premises and customer-controlled deployment; multi-tenant SaaS only after isolation and access control are proven.

> **A NOTE ON CRYPTO-AGILITY**
>
> Crypto-agility does not mean algorithms can always be swapped without code changes. Applications, certificates, protocols and hardware each carry constraints, and our platform makes them visible so they can be planned for.

SECTION 09

Four-month delivery plan

One hundred and twenty days from kick-off to a pilot-ready platform, with a clear exit gate at every stage.

PERIOD	DELIVERABLE	EXIT GATE
Days 1–10	Scope, lab, architecture, threat model, repository, backlog and standards register	Owners named; scope approved; lab reproducible
Days 11–30	TLS/certificate collector, schema, inventory and basic report	Correct findings on known lab assets
Days 31–45	Source/dependency scan, configuration import, risk rules, export and user workflow	Coverage matrix and tests reviewed
Days 46–60	Scanner MVP, installation guide and repeatable demo	Acceptance criteria met; limitations documented
Days 61–75	CBOM/dependency view, migration planner, one protocol or library integration	Plan traces back to evidence and dependencies
Days 76–90	Controlled PQC/hybrid test, performance and rollback tests	Tests pass for named configuration; limits recorded
Days 91–120	Pilot package, independent review, customer feedback, pricing and investor demo	Coherent demo and pilot-ready package

SECTION 10

Team, governance and operating rhythm

Clear ownership, short feedback loops and working software every week.

Capabilities and ownership

CAPABILITY	OWNERSHIP	RESPONSIBILITIES
PQC architecture and applied cryptography	CISO coordinates; engineers implement	Standards interpretation, library selection, protocol design, test strategy and technical review
Backend and platform engineering	Software engineering team	API, inventory, schema, job orchestration, rules and migration workflows
Discovery and security engineering	Cybersecurity team	TLS, certificate, source/dependency and configuration collectors
DevSecOps and integration	DevOps and engineering	Packaging, CI/CD, connectors, deployment and secure release
Frontend and product engineering	Shared across the team	Dashboard, reports and workflow
QA and security testing	Shared; interns support bounded tests	Ground-truth, regression, interoperability, performance and rollback testing
Product ownership and architecture	CISO, aligned with the founder's roadmap	Requirements, prioritisation, design-partner feedback and roadmap

The CISO coordinates the PQC architecture and discovery-scanner workstreams, drawing on our cybersecurity, software engineering and DevOps capability. Interns contribute to bounded, reviewed tasks such as test fixtures, documentation, lab environments, dependency inventories and regression automation. Cryptographic design decisions, security sign-off and production migration decisions remain with experienced team members.

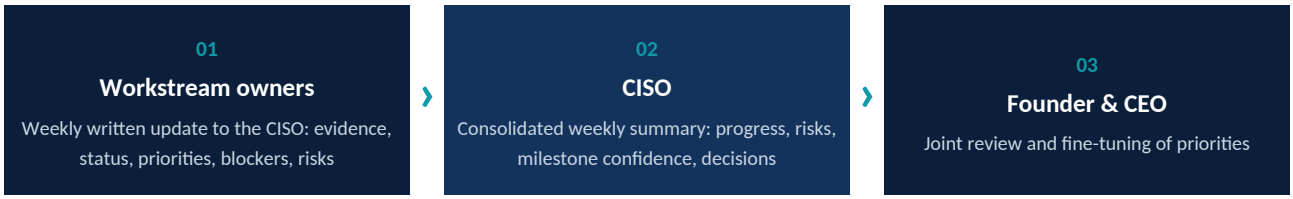
Every workstream has one accountable owner, a reviewer and a written definition of done. The roadmap is planned around our existing team; when capacity is tight, we narrow scope, rely on maintained standards-compliant libraries and propose sequencing trade-offs for decision, rather than pausing delivery.

Operating rhythm

MEETING / ARTEFACT	FREQUENCY	REQUIRED CONTENT
Engineering demo	Weekly	Working software and test results, not status-only presentations
Blocker review	Weekly	Top blockers, decisions needed, owner and resolution date
Product backlog	Continuous; reviewed weekly	Prioritised tasks, acceptance criteria, estimates and owners

MEETING / ARTEFACT	FREQUENCY	REQUIRED CONTENT
Standards register	Monthly and before releases	Versions, errata, supported algorithms and libraries, date checked
Security review	Each major milestone	Threats, permissions, data handling, dependencies and open findings
Founder/CEO progress note	Weekly, one page	Shipped work, tests passed, blockers, customer feedback, spend and decisions needed

How progress flows



The purpose is to surface issues early and help the team succeed, not to create reporting overhead. Changes to scope, sequence or architecture are documented with the reason, impact and decision owner.

HOW WE HANDLE SLIPPAGE

We run weekly demos. If a milestone slips, we identify the blocking dependency, propose options and revise scope openly. We never hide a delay or quietly reduce testing.

DEFINITION OF DONE

A task is complete only when code or documentation is reviewed, tests pass, security implications are considered, the feature is demonstrated and known limitations are documented.

SECTION 11

Security, safety and quality principles

A security company is judged first by how it secures itself. These principles are non-negotiable.

Authorised scanning only

We scan only systems we own or are explicitly authorised in writing to test, with defined targets, timing, permitted tests, contacts and stop conditions.

No customer secrets

Discovery never collects customer private keys or secrets. Any sensitive workflow requires redesign or explicit security and legal approval.

Data minimisation

We collect only what we need and document how it is stored, accessed and deleted.

Credential protection

Least privilege throughout; scan credentials protected and secrets kept out of logs and source code.

Complete audit trails

Every scan job, user action, configuration change, approval and migration result is recorded.

No unattended production change

No automated production cryptographic changes in the MVP: explicit approval, a maintenance window, validation and rollback are required.

Secure supply chain

Pinned dependencies, tracked licences and vulnerabilities, signed release artefacts and a maintained software bill of materials.

Expert review

Peer review of all cryptographic integration code; AI-generated implementation is never relied on without expert review and tests.

Honest claims

No claims of certification, guaranteed quantum safety, complete discovery or regulatory compliance without evidence and formal assessment.

Pilot-ready governance

Data-processing terms, permissions, incident response, vulnerability disclosure and support boundaries defined before every pilot.

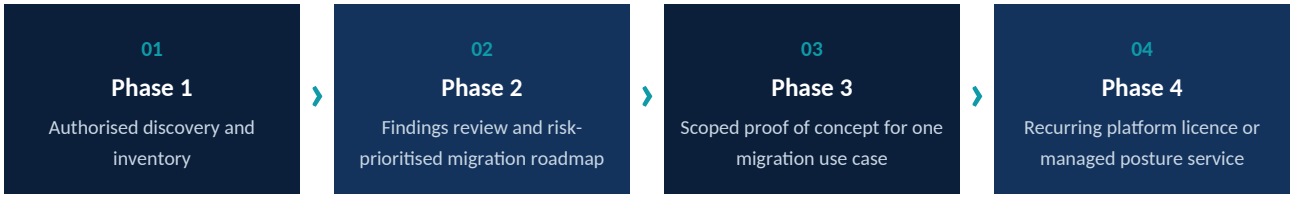
SECTION 12

Go-to-market and customer validation

Win one beachhead decisively, then expand with evidence.

Our priority segments are banking and financial services, telecommunications, data centres, healthcare, government and critical infrastructure. Rather than selling to every sector at once, we will select one beachhead and build depth, references and repeatability there first.

Entry offer: the Quantum Flow Cryptographic Readiness Assessment



Phase 4 begins once the product delivers repeatable value. Each phase is scoped and contracted separately, creating a natural path from assessment to long-term platform revenue.

Validating with customers

For every prospect, we capture the problem, current process, existing tools, decision-maker, budget owner, data-handling constraints, integrations, success criteria and willingness to pay. We track the true stage of every opportunity: a positive meeting is a conversation, not a commitment.

> **MONTH-FOUR COMMERCIAL GOAL**
>
> A credible demonstration, clear documentation, serious pilot discussions and a repeatable pricing approach. Any free pilot carries a written scope, access limits, success criteria and a path to a commercial decision.

SECTION 13

National Quantum Mission, ecosystem and funding

India's quantum-safe agenda creates real opportunity for indigenous IP, and we will pursue it with rigour.

The Department of Science and Technology's February 2026 report, "Implementation of Quantum Safe Ecosystem in India", is an important policy reference covering India's quantum-safe transition, PQC migration, testing and certification infrastructure and the adoption of quantum-safe solutions. We treat it as policy direction to translate into product requirements, not as a ready-made software kit.

Our approach to the ecosystem and grant funding

1. Prepare a three-to-five-page internal summary of the DST report relevant to discovery, migration, testing, interoperability and Indian IP.
2. Confirm current incorporation, DPIIT recognition, Indian ownership and IP-residency requirements for each specific programme.
3. Prepare a concept note: the problem, specific R&D, novelty, standards and open-source components, milestones, budget, team, IP ownership and validation environment.
4. Engage the relevant NQM Thematic Hub to confirm fit with an active call, dates, eligible expenses, co-development expectations and criteria.
5. Explore Karnataka start-up programmes and current MeitY, DST and technology-hub calls, each assessed on its own terms.
6. Secure one or two design partners; customer evidence strengthens both grant applications and private fundraising.

> **OUR DISCIPLINE ON FUNDING**
>
> We do not assume a grant is available because we operate in PQC. Eligibility, active calls and award decisions are confirmed in writing, and we describe our maturity precisely: prototype, pilot, product and certified solution are different stages.

OFFICIAL LINKS

DST report: Implementation of Quantum Safe Ecosystem in India
National Quantum Mission
NQM guidelines to support start-ups

SECTION 14

IP ownership and company readiness

Clean, well-documented IP is what turns engineering effort into enterprise value.

- Employees, contractors and interns sign appropriate confidentiality and IP-assignment terms before contributing code or designs.
- Authorship, creation date, third-party libraries and licences are recorded for every component.
- A software bill of materials is maintained, and open-source licences are reviewed before distribution.
- Product name, repositories, documentation, build pipelines and cloud accounts remain under company-controlled ownership.
- Original code and workflows are kept separate from partner code and licensed components.
- Architecture, threat models, tests, performance results and customer feedback are documented as evidence of product development.
- Before fundraising, we obtain legal advice on IP, open-source obligations, customer data, government procurement, security claims and patentability.

SECTION 15

Ambition, growth and our people

Ambition sets the direction. Evidence earns the outcome.

US$1B
valuation ambition by March–April 2027: a stretch goal, not a forecast

10+
countries targeted for presence by 2027

2
global investors/funds committed, subject to conditions

Valuation and investment

Our strategic ambition is to build Quantum Flow into a billion-dollar quantum-safe cybersecurity company, built in India and designed for the world. Key groundwork is in place: commitments from two global investors/funds, with funding to proceed in line with their agreed conditions and definitive documentation. The next critical milestone is a demonstrable beta and pilot-deployable scanner and migration platform. Product evidence, customer pilots, governance and commercial traction will be essential to support valuation.

International growth

The need to understand cryptographic exposure and plan a safe transition to quantum-resistant technology exists in every market. Our platform is designed for repeatable deployment, with country-specific regulatory, infrastructure and customer requirements handled deliberately. We are already in discussions across the UAE, Vietnam, the United States and Europe, and we track the stage and next action of every opportunity with precision.

Reference customers and visibility

Strategic customers, including the Government of Karnataka and major enterprises, can provide the evidence and reference cases for wider adoption. Based on the confirmation received from the Government of Karnataka, we are working towards adoption of the platform as a national reference architecture, which the team will support with reliable engineering, evidence, documentation and delivery. We will earn recognition across print and digital media, industry events and major business forums through results, not unsupported claims.

Our people

The people building this capability now are gaining scarce, practical expertise in one of technology's most important emerging fields: cryptographic discovery, crypto-agility, PQC migration and secure deployment. As the company grows, contributions will receive appropriate recognition and fair compensation through transparent, properly documented arrangements. Any equity or employee participation will be offered through a legally documented plan with clear eligibility and vesting terms, never through informal promises. Our shared responsibility is to build the strongest possible foundation through execution.

SECTION 16

The first ten working days

Momentum starts now. These are the actions that set the next four months in motion.

1. **Organise the team.** The CISO identifies contributors, agrees responsibilities and runs one shared backlog with a weekly progress review.
2. **Agree the MVP** scope, supported environments and explicit non-goals.
3. **Summarise the standards:** NIST FIPS 203, 204 and 205, with current errata and related guidance.
4. **Summarise the DST report** on India's quantum-safe ecosystem for internal use.
5. **Map the market** from public sources, including QShield 2.0, without using proprietary material.
6. **Set up the foundations:** lab, Git repository, CI, issue tracker, coding standards and dependency and licence register.
7. **Define the core model:** the common cryptographic asset schema and first acceptance tests, before expanding scope.
8. **Build the backlog** with bounded tasks for experienced engineers and interns.
9. **Line up customer voice:** identify 8–12 customer and design-partner interviews with a consistent questionnaire.
10. **Prepare the NQM concept note,** verifying current eligibility and calls before a full application.
11. **Lock the rhythm:** weekly demos, team-to-CISO updates and CISO-to-CEO summaries; publish a 30-day milestone plan with named owners.

APPENDIX A

Minimum cryptographic asset record

The common record every collector and workflow in the platform shares.

FIELD	MEANING
Asset ID	Stable Quantum Flow identifier
Asset type	TLS endpoint, certificate, source finding, dependency, configuration item or key-management service
Source	Network observation, source scan, configuration import or customer inventory
Target / location	Hostname, application, repository, cloud resource or system identifier
Algorithm / protocol	Observed or inferred value, with evidence and confidence
Key / signature details	Key type and size, or signature algorithm, where available
Evidence	Timestamp, scan method, output and collector version
Confidence	Observed, inferred, imported or unknown, with rationale
Business context	Owner and criticality supplied by the customer
Dependencies	Applications, libraries, certificates, vendors or services that rely on the asset
Risk factors	Exposure, confidentiality lifetime, business impact, migration complexity and support status
Recommendation / status	Investigate, vendor upgrade, migration planned, migrated, exception or accepted risk
Audit fields	Scan run, timestamps, user action, reviewer and policy version

APPENDIX B

Glossary

Key terms in plain language.

TERM	SIMPLE EXPLANATION
PQC	Post-quantum cryptography: cryptography designed to resist attacks by both quantum and classical computers.
ML-KEM	NIST-standardised key-encapsulation mechanism used to establish a shared secret (FIPS 203).
ML-DSA	NIST-standardised digital-signature algorithm (FIPS 204).
SLH-DSA	NIST-standardised stateless hash-based digital-signature algorithm (FIPS 205).
KEM	Key-encapsulation mechanism: establishes a shared secret rather than directly encrypting application data.
CBOM	Cryptographic Bill of Materials: an inventory of cryptographic components, algorithms, dependencies and evidence.
Crypto-agility	The ability to change algorithms and configurations through controlled processes.
Hybrid deployment	A supported protocol configuration combining classical and post-quantum mechanisms, as defined by the implementation or standard.
PKI	Public Key Infrastructure: the systems that issue, manage and validate digital certificates and public keys.
Design partner	A potential customer working with the product team to validate requirements and test an early version.
Known-answer test	A test that checks an implementation against an expected result for a specified input.

APPENDIX C

References and quick links

NIST FIPS 203: ML-KEM
csrc.nist.gov/pubs/fips/203/final

NIST FIPS 204: ML-DSA
csrc.nist.gov/pubs/fips/204/final

NIST FIPS 205: SLH-DSA
csrc.nist.gov/pubs/fips/205/final

NIST Post-Quantum Cryptography project
csrc.nist.gov/projects/post-quantum-cryptography

NIST CSRC publications
csrc.nist.gov/publications

NIST SP 800-57 Part 1: key management
csrc.nist.gov/pubs/sp/800/57/pt1/r5/final

DST: Implementation of Quantum Safe Ecosystem in India
dst.gov.in/sites/default/files/Quantum-Safe-Ecosystem-in-India.pdf

National Quantum Mission
dst.gov.in/national-quantum-mission-nqm

NQM guidelines to support start-ups
dst.gov.in/sites/default/files/Guidelines%20to%20support%20start-ups.pdf

QNu Labs QShield 2.0 (public product page)
www.qnulabs.com/quantum-security-ai-platform/qshield-2-0

IMPORTANT NOTICE

This document sets out product direction and strategic intent. It is not a guarantee of technical performance, certification, grant eligibility, customer adoption, funding or valuation. Statements about investor commitments, public-sector engagement and market discussions reflect their current status; applicable conditions, definitive documentation and approvals are managed through the appropriate company processes.

Official standards and guidance are our primary references, and we record assumptions and limitations openly. This document is confidential and intended for the Quantum Flow team, partners and prospective investors.

QUANTUM FLOW TECHNOLOGIES

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